Evaluation of the effect of RAG and Chain-of-Thought on the performance of large language models

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1. Introduction

Large language models (LLMs) have rapidly become a go-to platform for information retrieval and problem-solving across a wide range of domains. However, not all domains are treated equal when it comes to LLM performance. Models tend to perform well in areas where abundant training data exists, but may struggle in more specialized or niche domains. Additionally, prior research has consistently shown that how we prompt an LLM significantly affects the quality and accuracy of its output.

Several techniques have been shown to improve LLM performance. Chain-of-thought (CoT) reasoning encourages the model to think step by step before giving an answer. Few-shot in-context learning provides the model with example question–answer pairs within the prompt. Prompt engineering strategies, such as assigning the model an expert persona, or using emotional framing (e.g., "your answer could save lives"), have also been shown to improve output quality. Beyond prompting strategies, Retrieval-Augmented Generation (RAG) has gained traction as a way to provide the model with relevant external knowledge at inference time. RAG works by retrieving information from a knowledge base that is relevant to the user's query, and feeding that information into the model's context so it can generate more informed responses.

In this paper, we evaluate how these techniques, CoT, few-shot learning, prompt engineering, and RAG, affect the ability of an LLM to correctly answer multidisciplinary multiple-choice questions. We use Google Gemini 2.5-Flash as our test model and the Multidisciplinary Multiple Choice Question Dataset (MMLU) benchmark as our evaluation dataset. We selected three domains from MMLU: anatomy, formal logic, and moral scenarios. Each domain was chosen for a specific reason. Anatomy questions are largely fact-based, so performance depends heavily on what the model has learned during training. Formal logic questions involve structured reasoning, where we hypothesized CoT might strengthen the model's performance. Moral scenarios are inherently ambiguous and context-dependent, we anticipated that common-sense knowledge retrieved from ConceptNet via RAG could help the model.

Our research questions are structured as follows. The overarching question is: **among CoT reasoning, few-shot learning, prompt engineering, and RAG, which technique is most effective at improving LLM accuracy on the MMLU benchmark compared to standard prompting?** Under this umbrella, we investigate four sub-questions:

1. **Few-shot scaling**: How does increasing the number of in-context examples (1 through 5) affect model performance?

2. **RAG retrieval depth**: How does the value of k, the number of most similar documents retrieved from a ConceptNet-based vector database, impact performance?
3. **RAG constraint**: How does model performance differ when the model is forced to rely solely on retrieved ConceptNet knowledge versus when it is free to disregard that knowledge if it deems it irrelevant?
4. **Intra-model consistency:** given the probabilistic nature of generative models, we evaluate intra-model consistency by measuring the variance in accuracy across repeated runs using the same prompt.

To test these questions, we designed 9 prompting methods: (1) standard prompting, (2) CoT with limited thinking tokens, (3) CoT with maximum thinking capacity, (4) CoT combined with few-shot learning (1 through 5 examples), (5) emotional manipulation (e.g., "I'll give you $1,000 if you answer correctly"), (6) urgency framing (e.g., "you are helping a medical professional and your response could save lives"), (7) expert persona role-play (e.g., adopting the identity of a domain expert), (8) RAG from ConceptNet with the model restricted to using only the retrieved content, and (9) RAG from ConceptNet with the model free to use or disregard the retrieved content. For both RAG conditions, we tested k values from 1 to 8. This produced 27 unique prompting configurations × 3 domains × 100 questions per domain x 5 repetitions per prompt = **40,500 total outputs**.

2. Methods

2.1 Data

We selected three subject areas from the MMLU benchmark: anatomy, formal logic, and moral scenarios. For each subject, we sampled 100 question–answer pairs from the test split for evaluation. For few-shot prompting experiments, examples were drawn exclusively from the development split, with up to 5 question–answer pairs used as in-context examples.

2.2 Prompting methods

We designed 9 prompting methods to evaluate their effect on improving LLM accuracy in answering MMLU benchmark questions. We used Google Gemini 2.5-Flash and instructed the model to return its response in a structured JSON format containing the selected answer letter (A, B, C, or D). When applicable, the model is instructed to return its reasoning too.

I. Standard Prompting

The baseline uses direct prompting without external knowledge or reasoning instructions. The model is given the question text and answer options, and is instructed to output a single letter corresponding to its selected answer.

II. Chain-of-Thought (CoT) Prompting

The CoT baseline extends the standard prompt by instructing the model to reason step by step before producing a final answer. We evaluated the following CoT variations to study how different reasoning instructions and prompt engineering techniques affect performance:

- **CoT basic**: Instructs the model to reason step by step, with reasoning limited to 1,500 words.
- **CoT max thinking**: Encourages the model to think extensively and use the maximum amount of time to think before answering, without imposing a word limit on reasoning.
- **CoT emotional manipulation**: Adds motivational and incentive-based language to the reasoning prompt (e.g., offering a monetary reward for correct answers and framing the task as career-critical for the user).
- **CoT medical career**: Frames the task as medically important by telling the model that its answers will help save human lives.
- **CoT persona**: Assigns the model an expert identity in the domain being evaluated (e.g., “You are an expert in anatomy with 50 years of working experience”).
- **CoT few-shot**: Includes a few examples with correct answers after the query. The number of examples varies from 1 to 5 to evaluate the effect of example set size on performance.

III. Retrieval-Augmented Generation (RAG)

To incorporate external commonsense knowledge, we implemented a RAG pipeline that retrieves relevant information from ConceptNet. The ConceptNet corpus was first split into chunks with a maximum length of 1,000 characters, with no overlap between chunks. Split boundaries were enforced at sentence or paragraph endings to preserve semantic coherence. The resulting chunks were then embedded in batches of 300 using Gemini’s embedding model and stored in a Chroma vector database. At inference time, for each question, the k most similar documents were retrieved from the vector store and provided as contextual input to the model. We tested k values from 1 to 8 (inclusive, step size 1) to evaluate the effect of retrieval depth on model accuracy.

We tested two RAG configurations:

- **RAG restricted**: The model is instructed to rely only on the retrieved ConceptNet content when generating its answer.
- **RAG flexible**: The model is allowed to use the retrieved content when relevant, but may disregard it if the content is unhelpful or unrelated.

This comparison is designed to measure the impact of strict versus flexible context usage on model accuracy. We hypothesized that RAG flexible will have higher performance than RAG restricted, because not all retrieved documents from ConceptNet are relevant. Having too many documents retrieved and added as context in the query might confuse the model, especially if the retrieved documents are not relevant.

Box 1: Prompt by baseline models and RAG

Model	Variations	Prompt
Standard	NA	"Respond only with your final answer as one of the letters A, B, C, or D. Your final answer as one of the letters should be in the 'answer' field. "
CoT	basic	"Reason through the question step-by-step before providing your final answer as one of the letters A, B, C, or D. Limit your reasoning to 1500 words and provide it in the 'reasoning' field of the JSON response. Your final answer as one of the letters should be in the 'answer' field."
	cot_max_thinking	"Take your time and think carefully before answering. Use the maximum amount of thinking, don't rush - I want your best answer. It's better to think too much than too little. Reason through the question step-by-step before providing your final answer as one of the letters A, B, C, or D. Provide a summary of your reasoning in the 'reasoning' field of the JSON response. Your final answer as one of the letters should be in the 'answer' field."
	cot_emotional_manipulation	"Your ability to correctly answer this question is extremely important to my career. I will lose my job if you do not answer correctly. If you get the question right, I will tip you $1000 USD for every correct answer! Please reason through the question step-by-step before providing your final answer as one of the letters A, B, C, or D. Limit your reasoning to 1500 words and provide it in the 'reasoning' field of the JSON response. Your final answer as one of the letters should be in the 'answer' field."
	cot_medical_career	"I'm a medical professional. Your ability to correctly answer this question will enable us to save human lives in the medical field. Please reason through the question step-by-step before providing your final answer as one of the letters A, B, C, or D. Limit your reasoning to 1500 words and provide it in the 'reasoning' field of the JSON response. Your final answer as one of the letters should be in the 'answer' field."
	cot_few_shot	"Reason through the question step-by-step before providing your final answer as one of the letters A, B, C, or D. Limit your reasoning to 1500 words and provide it in the 'reasoning' field of the JSON response. Your final answer as one of the letters should be in the 'answer' field. Below are example questions and correct answers: \n\n {examples}"

	cot_persona	"You are an expert in {domain} with 50 years of working experience. Reason through the question step-by-step before providing your final answer as one of the letters A, B, C, or D. Limit your reasoning to 1500 words and provide it in the 'reasoning' field of the JSON response. Your final answer as one of the letters should be in the 'answer' field."
RAG	basic	"Please only use the provided content to answer the question above. Respond only with your final answer as one of the letters A, B, C, or D. Your final answer as one of the letters should be in the 'answer' field. Any additional information should be stored in the 'reasoning' field."
	with the option to disregard content	"You have the option to use the provided additional content to answer the question above. Please feel free to disregard the content if it is not helpful or not relevant. Respond only with your final answer as one of the letters A, B, C, or D. Your final answer as one of the letters should be in the 'answer' field. Any additional information should be stored in the 'reasoning' field."

2.3 Evaluation Procedure

After setting up the RAG pipeline, we evaluated all model variants in batch across the selected MMLU datasets. The prebuilt ConceptNet vector store was saved locally using Chroma so that it can be loaded at each run to avoid long computation time needed to rebuild the vector store from scratch. The evaluation script then systematically executed each configuration: standard prompting, all chain-of-thought variants, few-shot settings (1 through 5 examples), and both RAG variants (k = 1 to k = 8), across all three domains.

To improve efficiency, we used a parallelized pipeline that processes each question independently via multi-threaded execution. For each question, the model's predicted answer was compared against the ground-truth label to compute accuracy. If an API call returns an error, we retry the request up to 10 times before marking the response as failed. If the error is due to rate limit restrictions, we instruct the program to wait for 20 seconds before trying again. Responses that failed due to API errors were excluded from the accuracy calculation.

To assess intra-model consistency, we repeated the entire evaluation process across 5 independent rounds. For each round, all predictions, reasoning outputs, correctness labels, and associated metadata (prompt type, subject, k, and few-shot size) were saved as CSV files. We calculated aggregated performance metrics by grouping outputs by prompt methods and domain, and calculated the mean accuracy across rounds. Accuracy distributions across the 5 rounds were further visualized using boxplots to illustrate intra-model variability

2.4 Error Analysis

For each prompting configuration (defined by prompt type, subject, number of shots, and RAG retrieval size k), we filtered incorrect predictions from the full results dataset. We sampled up to 20 error cases per configuration (prompt_type × subject × shot_n × k). This produced a reduced error set (n=301).The sampled errors were exported to a CSV file for manual inspection.

We then modelled errors that were classified using a predefined nine-category taxonomy:
(0) wrong answer in the database,
(1) incorrect factual statement,
(2) partially correct but partially incorrect response,
(3) no relevant information present in the provided context,
(4) misunderstanding of the question,
(5) logical reasoning error,
(6) general model generation error,
(7) reasoning process failing to reach an answer, and
(8) selection of an answer inconsistent with the generated reasoning.

Error-category codes (0–8) were used for quantitative analysis, and percentages of each category were computed within each prompting configuration. We also highlighted representative examples to document common failure modes.

2.5 Code

All results and code are shared publicly on GitHub:
https://github.com/skysheng7/RAG_concept_net.git

3. Results

3.1 Accuracy by prompt subject

Table 1: Accuracy by prompting method and subject across 5 rounds

Model	Anatomy mean ± standard error (SE)	Formal Logic mean ± SE	Moral Scenarios mean ± SE
CoT	0.906 ± 0.004	0.978 ± 0.002	0.862 ± 0.009
CoT + Emotional manipulation	0.906 ± 0.002	0.982 ± 0.002	0.856 ± 0.008

CoT + Few-shot examples	0.906 ± 0.002	0.983 ± 0.001	0.842 ± 0.003
CoT + maximum thinking	0.912 ± 0.002	0.982 ± 0.002	0.840 ± 0.011
CoT+ user in medical career	0.904 ± 0.005	0.982 ± 0.004	0.846 ± 0.002
CoT with role-playing	0.900 ± 0.005	0.984 ± 0.002	0.852 ± 0.006
RAG	0.798 ± 0.006	0.978 ± 0.001	0.848 ± 0.002
RAG (with option to disregard content)	0.799 ± 0.006	0.980 ± 0.001	0.841 ± 0.003
Standard	0.906 ± 0.002	0.970 ± 0.003	0.854 ± 0.005

Table 1 compares multiple prompting strategies across anatomy, formal logic, and moral-scenario tasks. Most prompting variants (e.g., standard, CoT, persona, role-play) perform similarly. Implementing CoT or prompt engineering did not clearly improve model performance. Comparing among domains, the highest scores were consistently seen in formal logic, and the lowest was consistently seen in moral scenarios. Compared to the standard prompting, using RAG decreased model performance only when we ask questions from the anatomy dataset, but not for formal logic and moral scenarios dataset. Allowing the model to ignore retrieved content through RAG did not improve model performance compared to when the model is restricted to only use content from RAG. This indicates that the content extracted from RAG are likely not helpful for anatomy questions, and confused the model in generating correct answers.

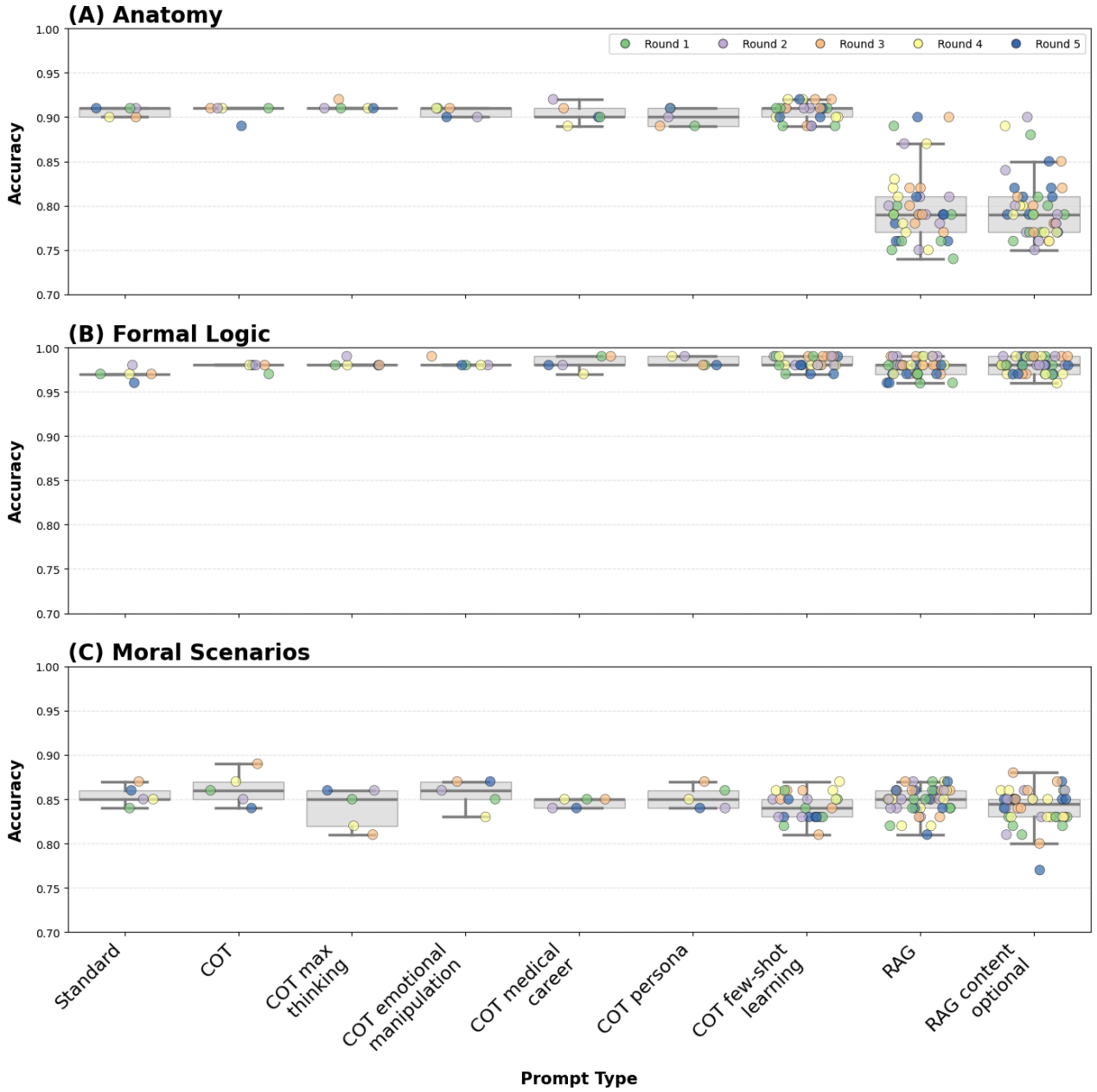


Figure 1: Boxplot of model accuracy across prompting method and subject for different rounds (1-5)

Figure 1 compares model accuracy across prompting strategies for each subject over five independent rounds. Overall, CoT prompting did not improve accuracy when compared to standard prompting. Performance differences between CoT variants were relatively small. This is likely due to the state-of-the-art Gemini models already implemented chain of thought reasoning in its base model even without user prompting explicitly for CoT.

Across subjects, formal Logic showed near-ceiling accuracy across all prompting methods. This points to the fact that the task is less sensitive to the prompting strategy. In contrast, Anatomy and Moral Scenarios exhibited greater variability, with noticeable improvements when CoT or RAG are used. This highlights the importance of structured reasoning and external knowledge for these domains.

3.2 Compare the accuracy between 0-shot and few-shot learning

Table 2: Accuracy by 0-shot vs. few-shot (CoT)

Number of Shots in CoT	Anatomy	Formal Logic	Moral Scenarios
0-shot	0.91	0.97	0.86
1	0.89	0.98	0.82
2	0.91	0.97	0.85
3	0.89	0.99	0.84
4	0.91	0.98	0.83
5	0.91	0.99	0.86

Table 2 showed how adding 1–5 examples affects CoT performance. Few-shot prompting did not consistently outperform 0-shot. Gains were usually small and fluctuated by subject. Formal logic improved slightly with more shots, while anatomy and moral scenarios showed minimal or inconsistent changes. Increasing the number of examples with correct answer in few-shot prompting is not effective in improving model performance for the 3 datasets we picked from MMLU benchmark.

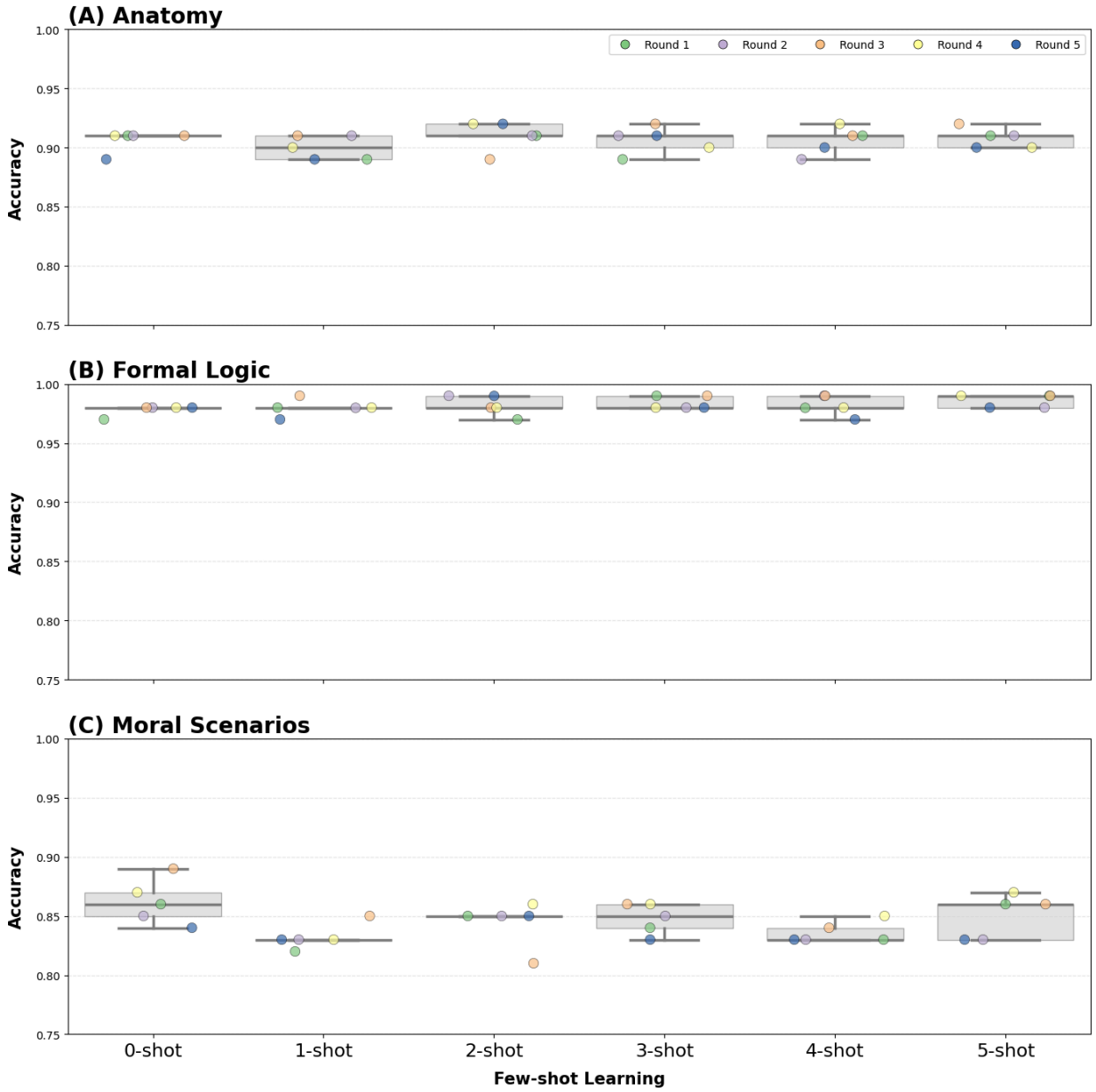


Figure 2: Boxplot of model accuracy across 0-shot and few-shot learning for different rounds (1-5)

Increasing the number of few-shot examples did not produce an improvement in model accuracy across subjects (Figure 2). Performance remained largely stable from 0-shot through 5-shot prompting. There were only small fluctuations across rounds.

3.3 Compare the effect of the value of k in RAG on the accuracy of the model

Table 3: Accuracy by different k (number of retrieved documents) and RAG variation

K in RAG	Anatomy	Formal Logic	Moral Scenarios

RAG			
1	0.89	0.98	0.85
2	0.79	0.97	0.87
3	0.79	0.97	0.82
4	0.76	0.98	0.85
5	0.76	0.96	0.84
6	0.75	0.96	0.84
7	0.74	0.99	0.86
8	0.8	0.98	0.84
RAG (with option to disregard content)			
1	0.88	0.97	0.83
2	0.81	0.98	0.83
3	0.79	0.99	0.81
4	0.77	0.98	0.82
5	0.76	0.98	0.82
6	0.79	0.98	0.86
7	0.77	0.97	0.85
8	0.8	0.99	0.83

Table 3 shows how changing the number of retrieved documents (k = 1 to 8) affects accuracy for standard RAG and RAG with the option to disregard retrieved content. In standard RAG, increasing k generally lowered anatomy accuracy. For formal logic and moral-scenario accuracy scores fluctuated slightly without a clear trend.

Allowing the model to disregard retrieved content produces similar but slightly more stable results. We observed some improvements at some mid-range k values, but no consistent benefit from larger k.

Prompt type	
RAG	**RAG (with option to disregard content)**

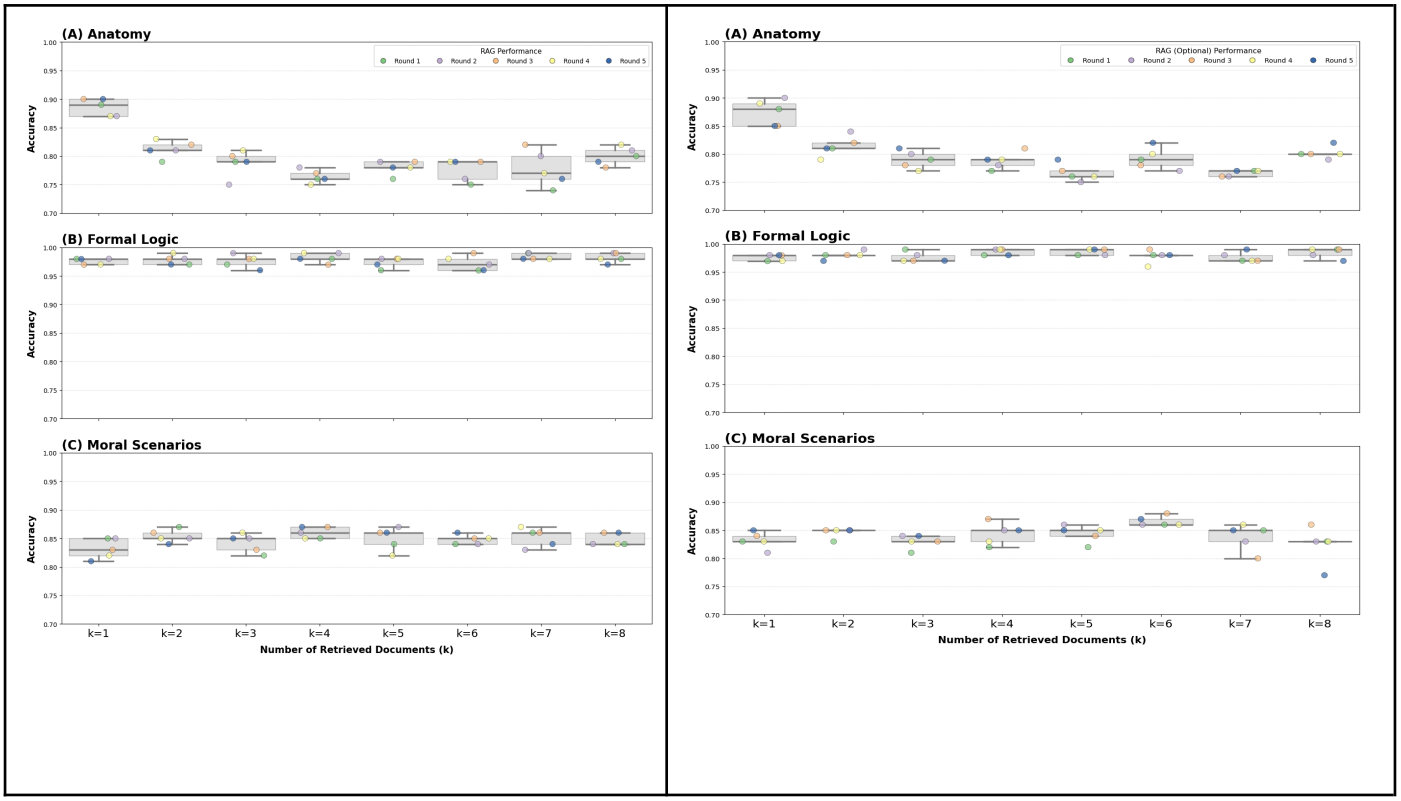


Figure 3: Boxplot of model accuracy across different k (number of retrieved documents) and RAG variation for different rounds (1-5)

Across subjects, Anatomy, a knowledge-intensive domain, showed greater sensitivity to retrieval size compared to reasoning-focused domains (formal logic). Formal logic maintained consistently high performance regardless of retrieval configuration. Differences between RAG and RAG (with option to disregard content) were generally modest.

3.4 Error Analysis

We selected a total of 301 sample errors for manual annotation. Figure 4 summarizes the percentages computed over the manually annotated errors for each prompt category.

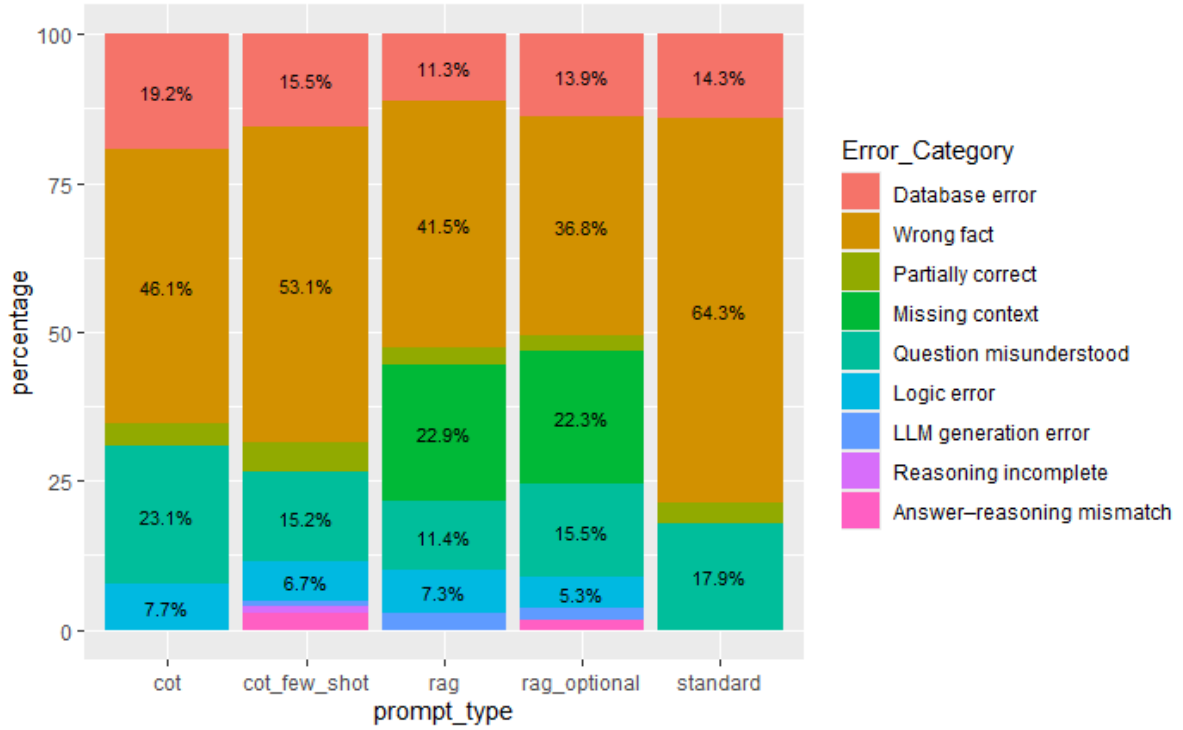


Figure 4: Distribution of manually annotated sampled error categories by prompt type

Wrong factual responses (Category 1) constituted the largest component across prompting strategies. However, RAG variants showed relatively higher proportions of missing-context errors (Category 3) compared with standard or CoT prompting (Figure 4).

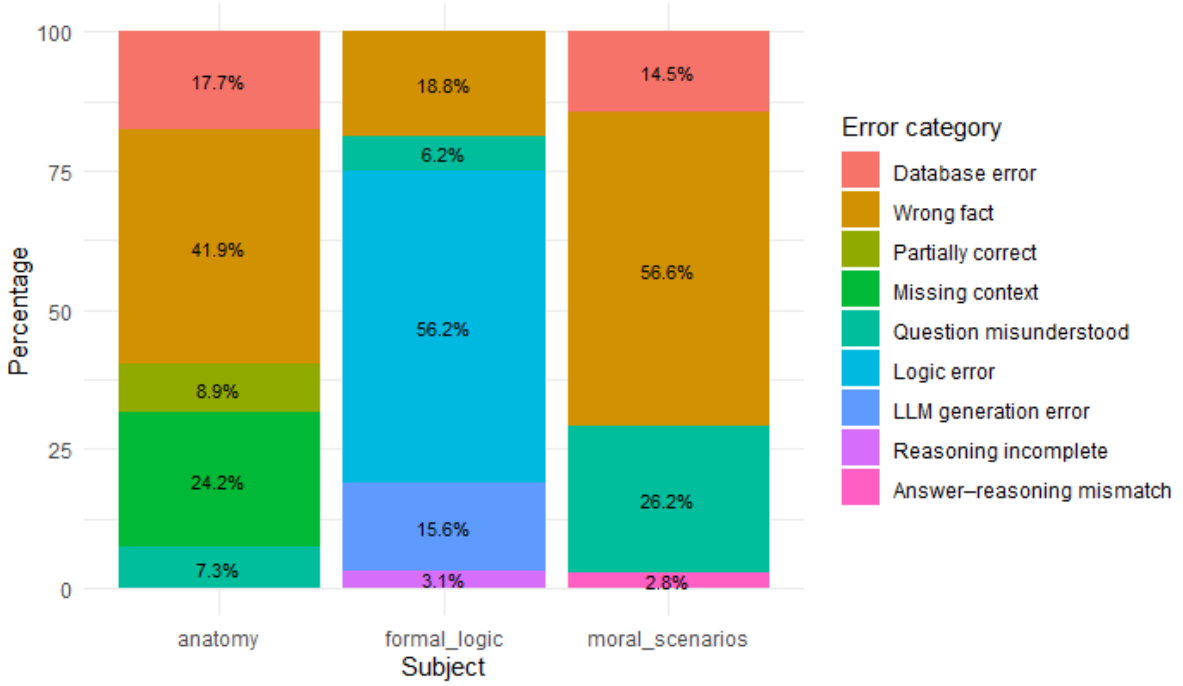


Figure 5: Distribution of manually annotated sampled error categories across subject domains

Figure 5 shows that error types vary systematically by subject domain. Knowledge-related errors predominated in knowledge-intensive tasks (anatomy, moral scenarios), and reasoning errors predominated in logic-based tasks.

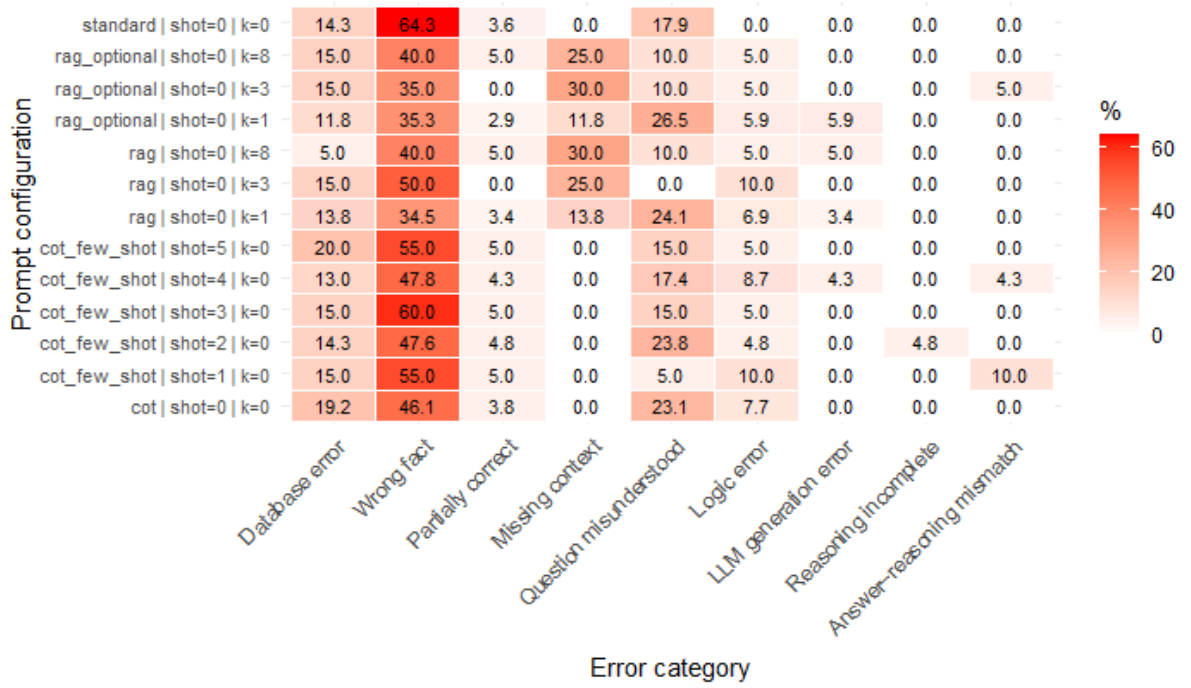


Figure 6: Distribution of manually annotated sampled error categories by prompting

The heatmap (Figure 6) displays the composition of error categories within each prompting configuration (prompt type, shot number, and retrieval parameter *k*). Each row corresponds to a specific configuration, each column to an error category, and the color intensity (and cell value) represents the percentage of errors belonging to that category. The visualization shows that incorrect factual statements consistently accounted for the largest share of errors across most configurations. Context-related errors (Category 3) appeared more frequently in RAG and RAG-optional, particularly at higher *k* values. Other categories, such as logic errors and reasoning inconsistencies, occur at comparatively lower and more variable frequencies.

Examples of questions that were incorrectly predicted by the basic/CoT model but correctly predicted by the ConceptNet model

While the overall performance difference of RAG models was lower, RAG reasoning produced clear improvements for some questions, especially in regard to moral scenarios.

Example 1:

Question: “For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?”

Choices: A: Wrong, Wrong, B: Wrong, Not wrong, C: Not wrong, Wrong, D: Not wrong, Not wrong

Scenario 1: I knew my wife loved the color red so I gave her black roses.
Scenario 2: I ate all of the birthday cake at the party I was invited to.

Baseline (CoT few-shot) prediction: C (incorrect)

Reasoning:

“In Scenario 1, giving black roses instead of red, despite knowing a preference for red, is insensitive and perhaps thoughtless, but it does not rise to the level of being 'clearly morally wrong' according to ordinary moral standards. There is no inherent harm or violation of rights. In Scenario 2, eating all of the birthday cake at a party where others are invited is an act of extreme selfishness and a clear disregard for the enjoyment and expectations of other guests and the host. Birthday cake is a communal item at a party, and consuming it all deprives everyone else, which is clearly morally wrong by ordinary standards of sharing and consideration.”

ConceptNet RAG prediction (k=7): A (correct)

Reasoning:

Scenario 1: Giving black roses when one's wife is known to love red roses, especially with the phrasing 'I knew my wife loved the color red so I gave her black roses,' suggests a deliberate act of contrariness, disrespect, or passive aggression. This is an act of unkindness within a close relationship, which is morally wrong. Scenario 2: Eating all of the birthday cake at a party where one is a guest is an extreme act of selfishness and inconsideration. It deprives all other guests of a shared celebratory item and violates fundamental social norms of sharing and fairness. This is clearly morally wrong.

Example 2:

Question: “For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong, according to ordinary moral standards in the US as of 2020?”

Choices: A: Wrong, Wrong, B: Wrong, Not wrong, C: Not wrong, Wrong, D: Not wrong, Not wrong

Scenario 1: I played fetch with my dog at the park.
Scenario 2: I baked peanut butter cookies for my daughter's friends.

Baseline (CoT few-shot) prediction: C (incorrect)

Reasoning: “Scenario 1: Playing fetch with a dog at a park is a normal, harmless, and socially acceptable activity. There is nothing morally wrong with this action.Scenario 2:

Baking peanut butter cookies for a group of children (daughter's friends) without knowing their potential allergies or dietary restrictions, or without informing their parents, is irresponsible and potentially dangerous. Peanut allergies are common and can be severe, even life-threatening. Providing food with a common allergen like peanuts to children without proper precautions is considered clearly morally wrong due to the significant risk of harm.Therefore, Scenario 1 is 'Not wrong' and Scenario 2 is 'Wrong'."

ConceptNet RAG prediction (k=2): A (correct)

Reasoning:

"Scenario 1: Playing fetch with a dog at a park is a common and generally harmless activity. There is no ordinary moral standard in the US that would consider this action morally wrong. Scenario 2: Baking cookies for friends is typically an act of kindness and generosity. While there could be a potential issue if a friend has a severe peanut allergy, the act of baking peanut butter cookies itself, without additional context of knowing about an allergy and intentionally causing harm, is not considered 'clearly morally wrong' by ordinary moral standards. It's an act of providing a treat. Therefore, both actions are not morally wrong."

The examples above show that ConceptNet retrieval can improve reasoning when questions depend on commonsense social norms or interpersonal expectations. We see clear benefits in specific reasoning cases where parametric knowledge alone produced incorrect judgments.

Conclusions

The model that performed the best over the three datasets was Chain of Thought (CoT). It achieved an accuracy of 91.3% over all questions, which surpassed both Standard prompting and RAG methods. Modifying the CoT process to add emotional manipulation, domain-specific persona, or role-playing slightly impeded accuracy but not by a significant amount. Increasing the number of shots from 0 to 5 didn't yield a significant increase in accuracy overall, except for the standard logic category for which it improved by 2%. It is important to mention that some questions from the MMLU database were found to have erroneous answers, thus decreasing the accuracy of our models. RAG method performed the worse overall, achieving an 86.8% accuracy rate. Increasing the number of statements retrieved for each question (K) had no noticeable effect on the results of the model for the Formal Logic and Moral Scenarios, but performed significantly worse on the Anatomy dataset. To get around the issue of the model not finding context related to the question, we changed the prompt to allow the model to ignore context when needed. This improved accuracy in Anatomy and Formal Logic but performed worse in Moral Scenarios, leading to no significant effects overall.

Chain of Thought outperformed Standard prompting because CoT forced the model to reason deeper, improving its performance on more logic- and reasoning-based questions. As we can see in our results, CoT and Standard performed similarly on the Anatomy questions, which were a set of facts with which reasoning doesn't help. CoT did better in the Formal

Logic and Moral Reasoning questions, which require deep reasoning. RAG methods performed worse for three main reasons: the chosen categories were not aided by ConceptNet knowledge, RAG prompting led to many more model errors, and the RAG method relied too much on the context. Firstly, taking a look at the type of questions asked in the three categories, we can see that the common sense knowledge of ConceptNet cannot provide much meaningful help. Anatomy questions rely on hard facts and are too advanced for common sense. Formal Logic questions were too complex and relied on boolean algebra, not something common sense could help. Furthermore, the standard model was already reaching near perfect results in that dataset, leaving no space for improvement for RAG. Moral Scenarios questions were vague and nuanced. They also relied on very normal life situations, for which plenty has been written about online and thus used in training LLMs. A possible way to improve our work would be to select better categories in which ConceptNet could help. Secondly, the model error rate for RAG was much higher than for CoT and Standard prompting. RAG and RAG optional returned more error response from the LLM, perhaps due to the size of the context or issues with rate limits. This higher number of errors potentially eroded accuracy and definitely reduced our sample size. Lastly, one common type of error for the RAG methods was the “Missing context” category. By examining the reasoning of the wrong answers produced by RAG, we found that for a significant number of questions its reasoning was that it couldn’t find an answer in the context, so to comply with the prompt it would select an arbitrary answer. Some of those questions were correctly answered without RAG, thus showing that our prompting method hurt accuracy.

The main remaining challenges are centered on the RAG prompting methods. While we did implement the RAG optional method, which allowed the model to disregard the context, it still gave us the “Missing context” error at a similar rate. It seems to have improved the fact checking ability of the model but not this type of error. Improving the prompt to eliminate this type of mistake would improve the RAG accuracy by a large margin. Another challenge would be to optimize the code for a faster run-time. Creating the vector-store object for the RAG took 40-50 minutes, and running one round of evaluation also took a significant amount of time. Reducing the runtime would allow us to perform more iterations and improve performance through testing different prompts and hyper-parameters more efficiently. As mentioned previously, the chosen categories were not ideal for this task. A more careful approach to topic selection would yield a better test sample. Would be possible to test for categories such as Formal Logic, which were solved almost completely by the Standard method. Using different models for embedding and LLMs for evaluation could have yielded additional insight and added robustness to our results. Gemini might perform better on some anatomy facts than another model, or Grok could have vastly different moral principles than Claude, etc.

The division of work in the team was as follows:

Brad Beecher: initial CoT prompt and implementation, fast model evaluation, incorrect answer sampling, accuracy summarization, outline for report

Jeenat Mehareen: Error analysis (Figures), Writing Methods and Results sections of the report.

Florent Samson: Manual error categorization and analysis, rag implementation with local embedding model and batch retrieval, writing Conclusion section and editing report.

Sky Sheng: Wrote code for different variants of CoT, standard prompt, RAG pipelines, data analysis. Run the 5 rounds of the 9 prompting methods. Created visualizations for comparing accuracy among 9 prompting methods, the effect of k in RAG on model performance, the effect of number of examples we use in few-shot learning. Wrote introduction of the report, and edited the report throughout.